

Finding the Right Balance: Relevance and Diversity in LLM Retrieval

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Abstract

Retrieval diversification is widely available in retrieval-augmented generation (RAG) frameworks, yet prior studies disagree on whether it improves retrieval and answer quality. We show that its effectiveness depends primarily on candidate-pool redundancy and the number of distinct evidence pieces required by a query. Using controlled near-duplicate injection and production-style overlapping chunking, we find that diversification harms relevance, evidence coverage and answer quality on clean pools, but becomes beneficial on multi-evidence tasks when redundancy causes nearest-neighbor retrieval to select repeated passages. We therefore introduce a query-adaptive rule that diversifies only when the effective number of distinct documents in the nearest-neighbor top- k selection falls below the query’s evidence requirement. Computed from existing embeddings, the rule captures most of the achievable gain, transfers across datasets and encoders and automatically reduces to nearest-neighbor retrieval for single-evidence queries. We also introduce RNG-SCORE, a geometric reranker with an exact nearest-neighbor fallback whose margin indicates duplicate structure. Overall, we conclude that diversification should be used selectively, based on observable redundancy and evidence requirements.

1 Introduction

Retrieval-Augmented Generation (RAG) augments a language model with external documents retrieved at inference time [5, 7, 25, 29, 40, 61]. In its simplest form, retrieval returns the top- k nearest documents to a query embedding. A long-standing concern with this strategy is redundancy: several retrieved passages may say nearly the same thing and thereby waste a context budget that language models already use unevenly [42]. This concern has motivated a growing line of work that injects an explicit diversity objective into the retrieval or reranking stage of RAG pipelines [17, 37, 41, 43, 47, 52, 65]. In the same spirit, diversification heuristics such as Maximal Marginal Relevance (MMR) [9] ship as built-in one-parameter retrieval modes in widely used frameworks,¹ so that a practitioner can toggle diversification on.

However, it is unclear whether diversifying document retrieval in RAG pipelines helps or harms. In fact, many works on the subject reach different conclusions. Diversity-aware pipelines report end-to-end gains of several points on multi-hop question answering [37, 52], while controlled evaluations of retrieval configurations find that MMR-style diversification yields no significant improvement over plain nearest-neighbor retrieval [6, 21].

These diverging conclusions are reached in different contexts where the retrieval diversity is given *instrumental* credit, meaning that diversity is not the end goal but rather a means to an end, as in a RAG pipeline. When an evaluation grants diversity *intrinsic* credit, namely when its metric suite rewards diversity directly, as the beyond-accuracy metrics of recommendation and search-result diversification do [16, 35, 69], then diversifying retrieval improves results by design.

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¹For example, `search_type="mmr"` in LangChain vector-store retrievers and `vector_store_query_mode="mmr"` in LlamaIndex. In both frameworks plain similarity search is the default and MMR is enabled by a single switch, with no guidance on when it should be.

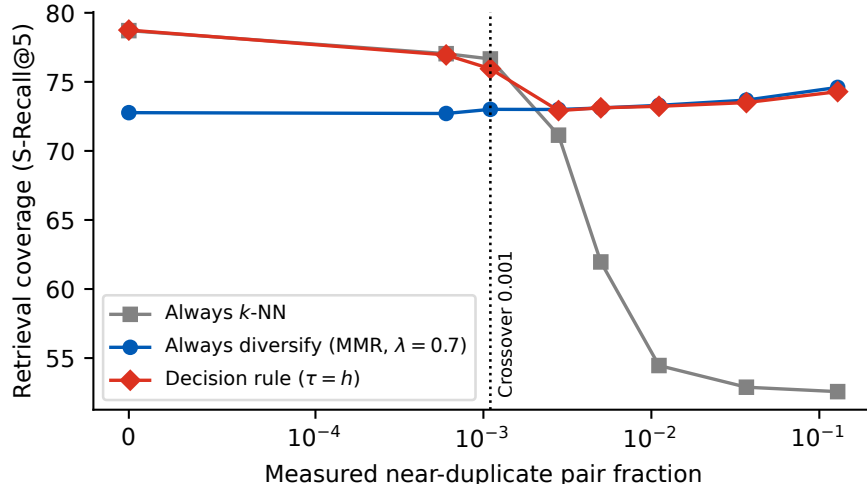


Figure 1: The paper’s central result: coverage of three policies on HotpotQA fullwiki as measured pool redundancy grows under controlled injection. Nearest-neighbor selection, the framework default, collapses past a measured redundancy of approximately 0.001. Diversifying hurts on clean pools and the decision rule of Section 4.3 tracks the upper envelope of both.

This paper analyzes the impact of diversity at the retrieval stage in a RAG pipeline. Thus, it is scoped to a context where diversity is granted instrumental credit. Within that scope we organize the question mainly around one observable variable, namely the redundancy of the candidate pool. This leads to a diagnostic computed from the system’s own embeddings which, thresholded at the query’s evidence requirement, suffices to decide whether to diversify. We also quantify how much of the available gain that rule captures. Figure 1 summarizes the result.

Furthermore, the study needs a diversifier predictable enough to serve as a stable probe across regimes. We use a simple geometric construction inspired by the concept of relative neighborhood graph (RNG) [67], namely the RNG-SCORE (see Figure 3). This score has provable do-no-harm behavior: at both extremes of its margin parameter γ it reduces exactly to nearest-neighbor ranking, so validated tuning can always fall back to pure relevance ranking. These properties make the RNG-SCORE a regret-bounded baseline whose validation-tuned margin additionally provides an offline readout of the pool’s duplicate structure.

The contributions of this paper include:

- (1) a *decision rule* which diversifies when the effective number of distinct documents in the nearest-neighbor top- k falls below the number of evidence pieces the query requires, does virtually no harm on clean pools, captures most of the achievable adaptive gain, transfers with no retuning across encoders, datasets, redundancy mechanisms and selection sizes and switches itself off on single-evidence retrieval where diversification has nothing to win;
- (2) a regime analysis of retrieval diversification organized around pool redundancy, with protocols that measure redundancy on the embedded pool and manipulate it directly (near-duplicate injection, production-style overlapping chunking, a weak-versus-strong pipeline contrast), all reported on one measured-redundancy axis;
- (3) a controlled, seed-replicated study of diversified reranking on fixed pools across multi-hop QA, open-domain QA and BEIR, showing that on clean pools every classical diversifier significantly harms both relevance and coverage-aware diversity;
- (4) per-query oracle bounds certifying that on clean pools, no query-adaptive policy can gain more than marginally, with headroom opening in proportion to measured redundancy;
- (5) misspecification regret, the worst case of a method over its hyperparameter range when the regime is unknown, as an evaluation criterion measured for every method; the RNG-SCORE’s regret is provably bounded and its tuned margin doubles as an offline diagnostic of duplicate structure.

The decision rule (1) is the most practical contribution: the crossover from the clean to the redundant regime is detectable per query, from the retrieval system’s own embeddings and the task’s evidence requirement, without relevance labels and without knowing which regime one is in.

2 Background and Related Work

2.1 Modern RAG pipelines

Modern RAG systems decompose retrieval into stages. A first-stage retriever produces a candidate pool by sparse, dense or hybrid retrieval [36, 53, 66, 76], typically accelerated with ANN infrastructure [33, 44]. The pool is usually refined before generation using stronger rerankers, cross-encoders or late-interaction models [2, 46, 55, 60]. Encoders range from compact sentence-embedding models [51] to stronger families such as BGE-M3 and Qwen3-Embedding [12, 75, 83] and proprietary services. Agentic, corrective, graph-based and chain-of-retrieval paradigms iterate the retrieval step [20, 62, 70, 79]. Diversification is orthogonal to all of these choices: every such pipeline ultimately selects k documents from a finite pool and the question is whether that selection should be diversified.

2.2 The case for diversity in RAG

Classical information retrieval offers several diversification mechanisms: MMR greedily trades relevance against similarity to already selected items [9]; DPP-based methods model repulsion probabilistically [13, 38, 74]; Maxmin and related dispersion objectives maximize spread directly [8, 24, 50]; and intent-aware approaches diversify over explicit query aspects [1, 56]. Moreover, work has been done to develop selective diversification in the context of web search results [57] based on specific characteristics of a given query, such as ambiguity [18].

A recent line of work argues that RAG specifically benefits from diversity-aware retrieval. Vendi-RAG [52] couples a Vendi-score retrieval objective [22] with an iterative pipeline in which a LLM judge scores candidate answers and the diversity weight is adapted across rounds. It reports accuracy gains of up to 4.2% on HotpotQA over Adaptive-RAG baselines [32]. DF-RAG [37] builds a geometric MMR variant and selects a per-query diversity level with planner and evaluator LLMs. It reports F1 gains of 4–10% over vanilla RAG with an estimated per-query oracle ceiling around 18%. Retriever portfolios extend the adaptive idea to the retriever level, training a lightweight router that selects per query among a small complementary set of retrievers including diversity-oriented members [63]. Submodular set functions have been applied to the reranking stage to balance relevance and coverage [17], alongside determinantal-point-process selection [41] and relevant-information-gain criteria [48], and scalable set-selection formulations have been proposed for the underlying combinatorial problem [43, 65]. Adjacent adaptive mechanisms let the pipeline decide how many documents to keep per query [45, 78] and ambiguous-question pipelines retrieve across inferred interpretations of the query [28]. A distinct, query-side notion of diversity decomposes queries by lexical diversity for finer-grained relevance assessment [84] or replaces inter-document dissimilarity altogether with coverage of a sub-query demand distribution [82]. Diversity has also been found beneficial for open-ended generation where comprehensiveness or answer diversity is itself the target [27, 54] and for long-context selection [72]. Notably, the most recent entries in this line motivate diversification explicitly by pool redundancy: AdaGReS [47] targets redundant and near-duplicate chunks that waste a token budget. Pool redundancy is also emerging as an evaluation variable in its own right: standard benchmarks measure far cleaner than the finance, legal and patent corpora on which strong retrievers degrade [14] and exact deduplication pays off only outside the clean academic regime [59], though neither line evaluates the selection step itself.

The experimental regimes behind these positive results deserve attention: Vendi-RAG entangles diversification with extra LLM calls and never isolates the selection step on a fixed pool, while DF-RAG retrieves from long contexts chunked into fixed-length windows, a pool redundant by construction, and its headline gains come from per-query adaptive selection of the diversity level rather than from fixed diversification. The set-selection and reranking methods [17, 41, 47, 48] do isolate the diversification step on a fixed candidate pool, but not in the regime studied here: they evaluate on diversity-focused or redundancy-heavy benchmarks—AdaGReS targets near-duplicate chunks by design—or with older first-stage encoders such as Contriever [30]. None isolates the diversification step on pools drawn from nonredundant corpora with state-of-the-art encoders, the configuration of most production vector-search deployments.

2.3 The case against diversity in RAG

A complementary literature reaches the opposite conclusion. Controlled evaluations find that MMR offers no significant advantage over naive nearest-neighbor retrieval [21], that reranking is among the highest-leverage components of a RAG pipeline [71] and, in a biomedical study holding the generator and embeddings fixed, that MMR “sacrifices answer relevancy for diversity” while the dense baseline lands within noise of the best configuration [6]. Studies of context composition show that passages semantically related to the query but not containing the answer are precisely the ones that harm the generator [3, 19] and geometric diversification, by design, substitutes exactly such passages. Analyses of context sufficiency [34] likewise indicate that generation needs coverage of the required information, which geometric spread does not measure.

2.4 Diversification in recommendation

Recommender systems are the setting in which diversity is given intrinsic evaluation credit through beyond-accuracy metrics that reward explicitly genre coverage and intra-list dissimilarity [35, 69]. The standard methodology is post-processing reranking: a relevance-optimized recommender produces a size- m candidate list per user and a greedy reranker selects $n < m$ items balancing relevance against diversity [9, 10, 74]. Controlled comparisons under this protocol consistently find that diversification pays with DPP delivering strong coverage gains at deployment scale [13, 74]. Search-result diversification reaches the same conclusion under intent-aware metrics [1, 16, 56].

When diversity is given intrinsic credit, the central question of this paper, namely whether diversification in the retrieval stage of a RAG pipeline helps or harms, does not arise since diversity is instrumental in this context. We therefore scope the study to instrumental credit regimes, where diversity can pay only through the final answer, and refer the reader to the literature above for the intrinsic credit regime.

2.5 Diversity evaluation metrics

Separating useful coverage from raw spread requires both families of diversity metrics. Annotation-free measures, average pairwise distance (APD) and the Vendi score [22], quantify geometric spread without reference to task structure. Intent-aware measures from the diversification literature, α -NDCG [16], ERR-IA [11] and Subtopic Recall [81], reward covering the distinct facets a query requires and penalize redundant coverage. We derive subtopics without extra annotation: for multi-hop QA, the distinct gold supporting documents of a question; for retrieval tasks with relevance judgements, the relevant-document set of the query. Disagreement between the two families is diagnostic: a method that raises APD while lowering α -NDCG produces spread without coverage. The same annotation-free statistics, computed on the embedded pool before any labels are consulted, also serve as the redundancy diagnostics of Section 4.

2.6 Relative neighborhood graphs

The *relative neighborhood graph* (RNG) was originally defined for finite planar point sets under the Euclidean distance [67] and extends naturally to any finite semimetric space [31] (see Appendix B for the definition of a semimetric). Two points x, y form an edge when no third point z satisfies both $d(z, x) < d(x, y)$ and $d(z, y) < d(x, y)$. Equivalently, the open lune of x and y , namely the intersection of the open balls of radius $d(x, y)$ centered at x and y , is empty (Figure 2).

In the retrieval setting, $x = q$ is the query and $y = w$ is a candidate document. A third document v with $d(q, v) < d(q, w)$ and $d(v, w) < d(q, w)$ lies in the lune of q and w : it is closer to the query than w and also close to w , making w geometrically redundant relative to q . Such a point v is said to *obstruct* the pair (q, w) .

RNG-style edge pruning is foundational in graph-based approximate nearest-neighbor indexing: HNSW, NSG and DiskANN sparsify neighborhood graphs by discarding edges whose lune contains another indexed point, at index-construction time and between corpus points, with DiskANN’s α -pruning relaxing the lune test by a multiplicative slack [23, 44, 64]. Recent work instead brings diversity into ANN retrieval, either by diversifying the graph search itself [73] or by designing data structures for diverse similarity queries under a hard diversity objective [4]. The present paper uses the same lune condition at query time, anchored at the query q as in Figure 3, but softens the binary test into a cumulative penalty that ranks a fixed pool rather than pruning a graph or enforcing a hard constraint. To our knowledge, query-centered soft relative-neighborhood scoring has not previously been studied as a diversification reranker.

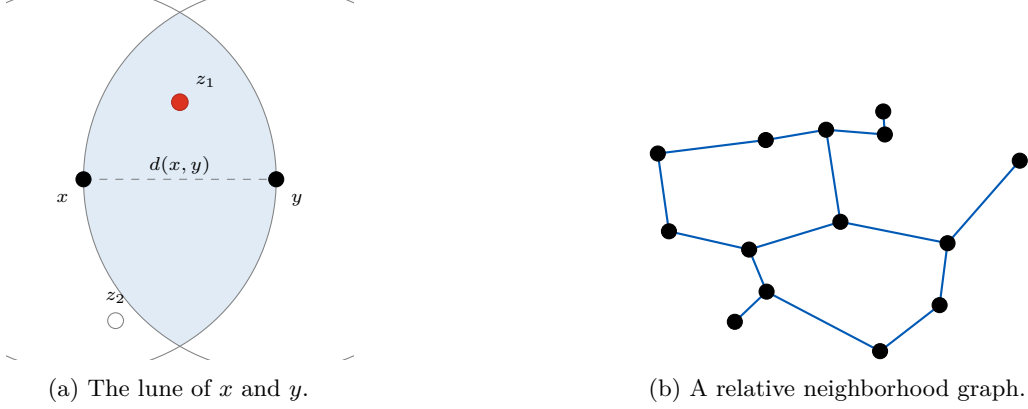


Figure 2: (a) The open lune of x and y (shaded) is the intersection of the two open balls of radius $d(x, y)$ centered at x and y ; z_1 lies inside it and obstructs the pair $\{x, y\}$, so that pair is not an edge of the RNG, while z_2 does not. (b) The RNG of a planar point set, with an edge between exactly the pairs whose lune is empty.

3 The Relative-Neighborhood Score

In order to analyze the impact of diversification at the retrieval stage of a RAG pipeline, we use different diversifiers such as MMR [9] and DPP [13, 38]. Here, we introduce the RNG-SCORE, a soft score used as a reranker and which can be tuned either to diversify retrieval or to fall back to k -NN as needed.

3.1 Setup

Let $V = \{v_1, \dots, v_m\} \subset \mathbb{R}^D$ be a finite candidate pool for a fixed query $q \in \mathbb{R}^D$. We assume the pool is the result of a first-stage retriever and that $m \gg k$ but remains moderate enough for pairwise reranking. Distances are denoted by $d(\cdot, \cdot)$. Throughout the paper, it is enough to assume that d is a semimetric on $V \cup \{q\}$.

For a candidate $w \in V$, define its set of *closer competitors*

$$P(w; q, V) = \{v \in V : d(q, v) < d(q, w)\}.$$

These are the only documents that can make w redundant relative to the query.

3.2 Definition

Definition 3.1 (RNG-SCORE). Let $\gamma \in \mathbb{R}$. For $w \in V$, define

$$\text{score}_\gamma(w; q, V) = d(q, w) + \sum_{v \in P(w; q, V)} [d(q, w) - d(v, w) + \gamma]_+, \quad (3.1)$$

where $[x]_+ = \max(x, 0)$.

The first term is simply the distance from w to the query, while each summand is an *obstruction penalty*. It is positive exactly when $d(v, w) < d(q, w) + \gamma$, i.e. when a closer document v sits close enough to w to make w look redundant from the viewpoint of q . Positive γ makes the penalty stricter, negative γ relaxes it. The induced reranker keeps the k lowest-scoring candidates with a computational cost of $O(m^2 D)$, which represents a fraction of a millisecond per query at $m = 100$ and negligible against embedding and generation, so reranking cost is not studied further.

Table 1 shows the three selectors on a real HotpotQA fullwiki query: k -NN fills the list with near-identical sibling articles and misses the second gold document, the RNG-SCORE keeps one cluster representative and recovers the second document while MMR disperses onto less relevant material.

3.3 Properties and instantiation

The score deviates from pure relevance only where there is geometric obstruction and is otherwise pinned to nearest-neighbor ranking, which is what lets its tuned parameter reflect the pool rather than the tuner. In

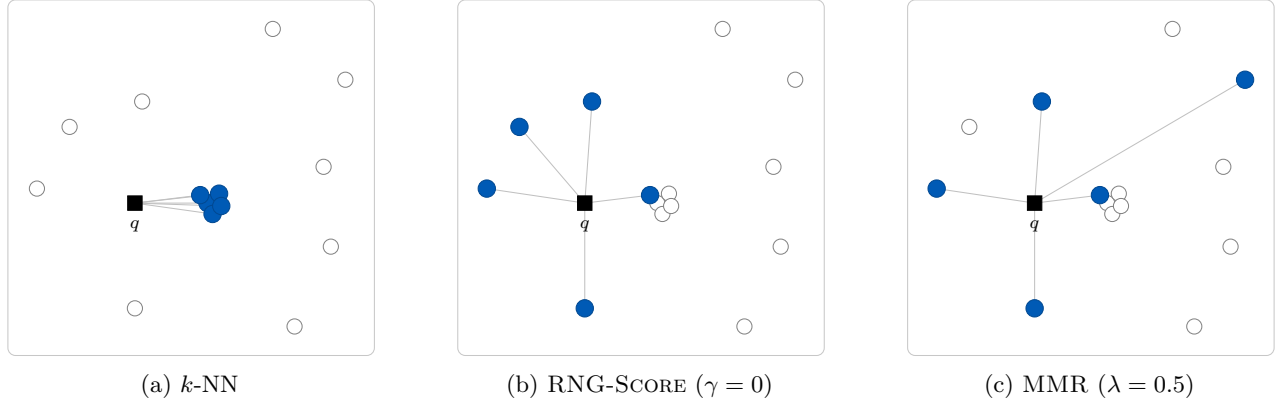


Figure 3: One candidate pool, three computed top-5 selections (filled blue = selected, open = unselected, $\blacksquare = q$). (a) k -NN spends the whole budget on a tight cluster of near-duplicates; (b) the RNG-SCORE keeps one cluster representative and disperses to nearby distinct points; (c) MMR pushes further out onto less relevant points.

Table 1: A HotpotQA fullwiki query needing two pieces of evidence (*1925 Birthday Honours* and *George V*, in bold).

rank	k -NN	RNG-Score ($\gamma=0$)	MMR ($\lambda=0.5$)
1	1925 Birthday Honours	1925 Birthday Honours	1925 Birthday Honours
2	1915 Birthday Honours	David Lloyd George	Edward VII
3	1924 Birthday Honours	1915 Birthday Honours	George Champagne
4	1926 Birthday Honours	Henry V, Holy Roman Emperor	Julius Meinl V
5	1927 Birthday Honours	George V	A. Thynn, Marquess of Bath

particular, it reduces to k -NN ranking at both extremes of the margin (Propositions A.4 and A.5), so a margin grid containing a deactivating margin ($\gamma \leq \gamma_0$ in Proposition A.4, under cosine distance $\gamma = -2$ always suffices) contains the relevance-only baseline as a reachable operating point (Corollary A.6). Validated tuning over such a grid never underperforms k -NN on the validation set and misspecification regret is bounded by how much an interior margin can perturb the ranking. No competing diversifier has this two-sided floor: Maxmin and DPP have no parameter to fix, while when increasing the parameter λ in the MMR algorithm, the obtained selection varies from pure diversity ($\lambda = 0$) to pure relevance ($\lambda = 1$).

In many RAG pipelines, documents in the corpus are normalized and similarity between documents v and w is measured using cosine similarity, which we note $\langle v, w \rangle$. In this context, the corresponding distance is $d(v, w) = 1 - \langle v, w \rangle$ and the RNG-SCORE for given query q and pool V becomes

$$\text{score}_\gamma^c(w; q, V) = 1 - \langle q, w \rangle + \sum_{\substack{v \in V \\ \langle q, v \rangle > \langle q, w \rangle}} [\langle v, w \rangle - \langle q, w \rangle + \gamma]_+. \quad (3.2)$$

Furthermore, the RNG-SCORE is adaptable to the context where the query–document distance differs from the distance in the embedding space, as is the case when using a cross-encoder (Appendix B). We may also adapt the RNG-SCORE so that it takes as input a kernel analogously to DPP (Appendix C).

4 Redundancy Diagnostics and the Decision Rule

The regime analysis of this paper is only actionable if the controlling variable can be observed by a deployed system cheaply and without labels. This section defines the two diagnostics we use, at the deployment level and at the query level, and the decision rule built on the latter. Both diagnostics are computed from the embedded candidate pool that the retrieval system already holds: neither consults relevance judgements, gold answers or any model beyond the existing encoder.

4.1 Near-duplicate pair fraction

For an embedded pool V with unit-normalized embeddings, define

$$\text{PoolRedundancy}(V) = \binom{|V|}{2}^{-1} |\{ \{v, w\} \subset V : \langle v, w \rangle > 0.95 \}|, \quad (4.1)$$

the fraction of passage pairs whose cosine similarity exceeds a near-duplicate threshold, fixed at 0.95 throughout. Two properties of the statistic should be kept in mind when comparing across deployments: the cutoff’s meaning depends on the encoder’s similarity scale and the pair fraction grows quadratically with duplicate-cluster size. Thus, comparisons should hold the encoder and the pool size m fixed, as all sweeps in this paper do. In our experiments V is each query’s top- m candidate pool and we report per-condition means over queries. Averaged over a sample of production queries, the same statistic places any deployment on the common measured-redundancy axis along which we report the synthetic injection sweep and the natural chunking sweep of Section 6.2, so that a measured value can be compared directly against the crossover observed in our experiments.

4.2 Effective size of the nearest-neighbor selection

The pair fraction (4.1) is a property of the whole pool. Nevertheless, what determines whether diversification helps for a specific query is whether redundancy crowds the top of that query’s ranking. We therefore use a sharper, selection-centered statistic. We run plain nearest-neighbor selection and measure the *effective number of distinct documents* among the k selected via the Vendi score [22] of their embedding similarity matrix:

$$T(q) = \text{Vendi}(k\text{-NN}(q)) \in [1, k]. \quad (4.2)$$

$T(q) \approx k$ means the k selected documents are mutually orthogonal, while $T(q) \approx 1$ means the selection has collapsed onto near-copies of a single document. The statistic costs one $k \times k$ eigendecomposition on embeddings already in memory, i.e. microseconds. An average-pairwise-distance variant of (4.2) behaves interchangeably in our experiments (Section 6.4).

4.3 The decision rule

The decision rule is a one-line policy with two frozen choices, a threshold $\tau \in [1, k]$ and a fallback diversifier D :

$$\text{select}(q) = \begin{cases} D(q) & \text{if } T(q) < \tau, \\ k\text{-NN}(q) & \text{otherwise.} \end{cases} \quad (4.3)$$

The threshold is set to the task’s evidence requirement, $\tau = h$, the number of distinct evidence pieces a query needs: two supporting documents give $h = 2$ on the multi-hop benchmarks, while $h = 1$ makes the trigger condition $T(q) < 1$ unsatisfiable, so on single-hop retrieval the rule reduces to k -NN by construction. The fallback diversifier D is selected on a validation set pooled across redundancy conditions and then frozen.

The rule is a realizable member of the family of query-adaptive policies that the all-methods oracle of Section 6.3 bounds from above, which makes the comparison meaningful in both directions: the oracle says how much any policy could gain, while the rule says how much of that a single-threshold policy actually captures. Section 6.4 confirms the anchoring empirically: threshold tuning left free recovers $\tau^* \approx h$ at both $k = 5$ and $k = 10$, with a wide plateau of equivalent thresholds around it.

Note that the paper thus carries two thresholds with different roles. The near-duplicate pair fraction is a statistic measured on a pool of m documents while the rule threshold τ lives on the individual query: it tests the selection of $k \ll m$ documents the system has just made. The two are linked through the rule trigger rate: as a pool’s measured redundancy passes a certain crossover (see for example Figure 1), the per-query statistic $T(q)$ collapses for most queries and the fraction of queries falling below τ jumps from under 16% to over 80% (Section 6.4). Thus, thresholding each query at τ implements the pool-level boundary automatically without ever measuring the pool redundancy. The validation-tuned RNG-SCORE margin of Section 3 supplies a third complementary readout, offline and at the workload level since validation queries are needed to compute it.

5 Experimental Setup

This section details the research questions, datasets, models, baselines, evaluation metrics and protocols. The results follow in Section 6.

5.1 Research questions

The evaluation is organized around five research questions.

- RQ1** *Diversification on standard benchmarks.* On pools retrieved from nonredundant corpora with state-of-the-art encoders, does any geometric diversifier improve results quality over nearest-neighbor selection?
- RQ2** *Manipulating pool redundancy.* Does raising pool redundancy create a regime in which diversification pays and what is the risk of committing to a method without knowing the regime?
- RQ3** *Oracle headroom.* How much could any query-adaptive diversification policy gain in each regime?
- RQ4** *The decision rule.* Can the diagnostic of Section 4.2 decide per query whether to diversify and how much of the oracle headroom of **RQ3** does the rule capture?
- RQ5** *Answer quality under redundancy.* Do the findings carry over to end-to-end generation quality in a redundant regime?

5.2 Datasets

HotpotQA (fullwiki) [80] is the primary multi-hop benchmark: each question needs two supporting paragraphs from distinct articles and candidate pools are the unique passages of the top-100 MDR chains of Xiong, Li, Iyer, Du, Lewis, Wang, Mehdad, Yih, Riedel, Kiela, and Oguz [77] (between 80–120 candidates per query, two gold passages). *2WikiMultiHopQA* [26] and *MuSiQue* [68] (two to four hops) vary hop count at fixed credit and redundancy regimes. Candidate pools for 2WikiMultiHopQA are the ten paragraphs of its distractor setting, the supporting paragraphs plus distractors. MuSiQue, restricted to its answerable subset, provides roughly twenty paragraphs per question (two to four of them supporting), which form its pools. *Natural Questions Open* [39] with DPR top-100 pools [36] is a single-hop, single-answer control. For the redundancy sweep, we collapse a query’s answer-containing passages to one subtopic, since DPR marks every such passage relevant (around 4.6 per query) and would otherwise misread as multi-evidence. *BEIR* [66] supplies heterogeneous graded-relevance tasks (SciFact, FiQA-2018 and TREC-COVID together with the argument-retrieval tasks ArguAna and Webis-Touché-2020), with each query’s relevant set as its subtopics. SciFact anchors the single-evidence end (92% of queries one relevant document), FiQA-2018 the multi-evidence middle (typically two to three relevant documents per query) and TREC-COVID the saturated extreme (mean 493).

5.3 Models

Because the analysis applies to any finite candidate pool, we evaluate along an encoder-capacity axis. The strong stack uses **bge-m3** [12] (1024-dimensional) and **Qwen3-Embedding-4B** [83] (2560-dimensional) as dense encoders and **bge-reranker-v2-m3**, the reranker released alongside BGE-M3 [12], as the cross-encoder for the multi-stage half of **RQ1**. The weak encoder **all-MiniLM-L6-v2** [51], representative of lightweight deployments, supplies a capacity contrast for **RQ2**. Encoder capacity is treated as a third route into the redundant regime, not as a robustness check. Downstream generation uses **flan-t5-base** [15] as the reader (4-beam decoding).

5.4 Baselines

We compare k -nearest-neighbor selection (k -NN) against four reranking baselines applied to the same candidate pool: MMR [9] with trade-off weight $\lambda \in \{0.3, 0.5, 0.7\}$, Maxmin dispersion [50], greedy DPP selection [13, 38] and the RNG-SCORE with margin grid $\gamma \in \{-0.5, -0.3, -0.1, -0.05, 0, 0.05, 0.1, 0.3, 0.5\}$.² Maxmin is

²These are the default grids: the redundancy and chunking sweeps extend both toward heavier diversification pressure (Section 5.6) and two single-run analyses (the clean-pool generation comparison of Section 6.1 and the ArguAna and Webis-Touché columns of Table 3) use a fine margin grid, $\gamma \in [-1, 1.25]$ in steps of 0.05.

query-agnostic and is included as a diversity stress-test rather than as a model of relevance-aware reranking. We refer to MMR, Maxmin and greedy DPP collectively as *fixed-pressure* diversifiers: each applies a diversification pressure set by its parameters, with no interior operating point that recovers relevance-only ranking.

5.5 Evaluation metrics

We report four metric families so that relevance, raw spread, genuine coverage and downstream utility can be distinguished. *Relevance*: Recall@ k and NDCG@ k against gold passages or graded judgements. *Annotation-free diversity*: average pairwise cosine distance (APD) and the Vendi score [22]. *Intent-aware diversity*: α -NDCG@ k [16] with redundancy parameter $\alpha = 0.5$ and Subtopic Recall (S-Recall@ k) [81], with subtopics derived as in Section 5.2. *Generation quality*: exact match (EM) and token-level F1 against reference answers. All relevance, intent-aware and generation metrics are reported on a 0–100 scale; APD and the Vendi score are reported on their natural scales. Recall@ k counts all gold passages of a query, including the roughly 5% that the first-stage pool fails to contain while S-Recall@ k counts retrievable subtopics only, which is why the two differ at the baseline.

Misspecification regret. If the value of diversification depends on a regime variable that practitioners do not usually measure, then the practically relevant property of a method is not its tuned performance in either regime but its worst case across regimes. Writing $U_r(M_\theta)$ for the test utility (S-Recall@5) of method M at operating point θ in regime r , we report

$$\text{regret}(M) = \max_r \max_\theta [U_r(k\text{-NN}) - U_r(M_\theta)], \quad (5.1)$$

the worst shortfall against the relevance-only default, over distinct redundancy regimes and over the method’s full hyperparameter grid (the sweep grids of Section 5.6). Regret is a maximum over the grid, so the values computed on the extended sweep grids upper-bound those of the narrower default grids. For k -NN, the reference is the best tuned method in each regime. This is the minimax-regret construction of statistical decision theory,³ with regimes as states of nature and (method, operating point) commitments as actions.

5.6 Protocol

For each query, a first-stage retriever produces a candidate pool of size m and every reranking method selects k documents from the same pool, so that all differences are attributable to the selection step. Unless stated otherwise, $m = 100$ and $k = 5$ for QA tasks while $m = 100$ and $k = 10$ for BEIR tasks. Hyperparameters (γ for the relative-neighborhood scores, λ for MMR) are selected on a held-out validation split of 20% of the queries and reported on the remaining test split.⁴ Unless identified as single-run, every sweep and comparison table is repeated under three seeds, the seed controlling both the injected near-duplicates and the validation/test split. On HotpotQA, each split is 1,481 validation and 5,924 test queries. The same frozen splits underlie **RQ1** and **RQ2–RQ4**, so that all derived analyses are computed from one set of per-query results.

Statistical reporting. Unless stated otherwise, every reported value is a mean over the three seeds and “significant” means the median across seeds of per-seed paired Wilcoxon signed-rank tests against k -NN falls below 0.05. Across-seed 95% confidence intervals scale with the query-set size: they are below 0.8 points for every relevance, coverage and generation metric in the tables computed on HotpotQA-scale query sets (the injection sweep and its derived analyses, the cross-encoder comparison), reach 1.1 points on the QA columns of the chunk-overlap table and up to 8 points on the 50–300-query BEIR sets. The captions restate the bound wherever it exceeds 0.8 points. The annotation-free diversity columns, which keep their natural scales, are up to 0.013 for APD and 0.06 for the unbounded Vendi score. To keep tables and captions readable we do not repeat these details: exceptions are flagged where they occur and single-run results are identified as such. The comparisons reported are pre-specified, each diversifier against the k -NN default on the objective metrics of its table (a single metric on the sweep tables, four on Table 2).

³Minimax regret in the sense of Savage [58], except that the reference policy is the k -NN default rather than the per-regime optimum. The practical question is how much worse than not diversifying a committed choice can turn out to be.

⁴In practice the negative end of the default margin grid realizes the fallback of Corollary A.6: at $\gamma = -0.5$ the selected sets are identical to k -NN’s on every query of the seed-replicated BEIR tasks and match k -NN to reporting precision on the clean HotpotQA pools, so validated tuning can express “do not diversify”. The sweep grid’s -0.3 end deviates from k -NN on about 1% of clean queries.

Cross-encoder pipeline. The multi-stage experiments retrieve the top- m passages with **bge-m3**, rescore them with **bge-reranker-v2-m3** and hand the rescored pool to the rerankers. Because the RNG-SCORE is defined through a distance while a cross-encoder returns only query-document relevance, these experiments use the semimetric construction of Appendix B, with cross-encoder relevance as the query-document distance and embedding distance between documents.

Manipulating and measuring redundancy. Injection augments each pool with a fraction ρ of sentence-shuffled near-duplicate copies that retain their source’s identity (so coverage metrics count them as redundant, not as new evidence) drawn half from gold passages and half uniformly to reflect the concentration of redundancy around relevant content. Pools are re-embedded after injection. Because copies share an identity, NDCG credits only the first retrieved copy and hyperparameters are tuned on S-Recall@ k (the $\alpha = 0.5$ gain of α -NDCG@ k would otherwise reward repeated copies). The sweeps use the operating-point grids $\lambda \in \{0.3, 0.5, 0.7, 0.9\}$ and $\gamma \in \{-0.3, -0.2, -0.1, -0.05, 0, 0.05, 0.1, 0.2, 0.3, 0.5, 1\}$, the upper ends extended beyond the default grids of Section 5.4 after early HotpotQA sweeps tuned to a grid boundary. The headline sweep refines the grid where the regime turns, $\rho \in \{0, 0.025, 0.05, 0.1, 0.15, 0.25, 0.5, 1\}$, on HotpotQA with **bge-m3** (all 7,405 queries), replicated with **Qwen3-Embedding-4B** and **all-MiniLM-L6-v2** (encoder-capacity axis) and on SciFact and FiQA-2018 ($k = 10$), the single- and multi-evidence ends of the BEIR tasks. Overlapping chunking instead re-chunks each pool passage into 60-word sliding windows at overlap $\in \{0, 0.25, 0.5, 0.75\}$, inheriting source identity with no text perturbed, the mechanism of production ingestion pipelines. Both protocols report the measured redundancy (4.1), placing synthetic and natural sweeps on one axis.

Rule and oracle. The decision rule (4.3) fixes $\tau = h$, where h is the number of evidence pieces a given query requires, and selects the fallback D^* by maximizing mean S-Recall@5 on validation queries pooled across all injection levels (never observing the level identity). As a check on the anchoring, the threshold is additionally tuned freely over $[1, k]$ jointly with D . The evidence requirement is $h = 2$ on the two-hop datasets, the per-query hop count on MuSiQue, $h = 1$ on the single-answer datasets and the query’s relevant-set size on the graded-relevance BEIR tasks. The frozen rule is then evaluated against always- k -NN, always- D^* , the per-level tuned best method and the all-methods oracle. The per-level tuning selects, on each injection level’s validation split, the best method and operating point for that level. Both oracles operate per query with ground-truth access: the per-method oracle [37] sweeps one method’s grid and records the best value achievable, and the all-methods oracle additionally chooses the method, upper-bounding any test-time adaptive policy.

Generation. Generation experiments attach the reader to the sweep, generating an answer from each method’s top- k at every level and scoring it with the generation metrics of Section 5.5. The decision rule’s answers are reconstructed per query without extra generation, returning the diversifier’s answer when the trigger fires and the k -NN answer otherwise.

6 Results

6.1 RQ1: Diversification on standard benchmarks

Table 2 reports controlled reranking on HotpotQA with **bge-m3** (test splits, three seeds, paired significance against k -NN). Every classical diversifier trades relevance for spread, losing between 0.7 and 33.0 S-Recall@5 points significantly in every seed. Crucially, the intent-aware metrics fall with relevance as APD and Vendi rise. The tuned RNG-SCORE sits on top of k -NN, nominally significant on every objective metric but immaterial in magnitude (+0.3 S-Recall). On these pools, the correct amount of diversification is approximately none.

The retrieval losses propagate to generated answers (EM/F1 columns of Table 2). On the clean HotpotQA pools, every fixed-pressure diversifier loses EM against k -NN, while the tuned RNG-SCORE sits 0.2 EM points above k -NN. Section 6.5 carries this into the redundant regime, where the picture inverts.

The single-stage behavior of 2WikiMultiHopQA, MuSiQue and the NQ-Open single-answer control tracks their measured pool redundancy (Appendix D): 2WikiMultiHopQA, whose natural pools measure below the crossover of Section 6.2, reproduces these conclusions, while MuSiQue’s distractor pools measure at the crossover fraction and mild diversification pays about two S-Recall points. Under the cross-encoder pipelines below, all four datasets lead to the same conclusion, i.e. diversification hurts and top- k selection wins.

Table 2: Controlled reranking on HotpotQA (MDR top-100 pools, **bge-m3**, $m = 100$, $k = 5$). The four left metrics are objectives (best in bold, second best underlined); APD and Vendi are descriptive spread statistics, not objectives; EM and F1 are from single generation runs (**flan-t5-base**, full test set).

Method	Recall@5	NDCG@5	α -NDCG@5	S-Recall@5	APD	Vendi	EM	F1
k -NN	<u>74.8</u>	<u>73.9</u>	<u>76.5</u>	<u>78.7</u>	0.450	3.08	<u>34.1</u>	<u>45.3</u>
MMR ($\lambda = 0.3$)	45.3	53.3	55.4	48.1	0.630	4.00	23.3	32.4
MMR ($\lambda = 0.5$)	55.6	60.3	62.5	58.6	0.574	3.73	25.7	35.3
MMR ($\lambda = 0.7$)	69.2	70.0	72.5	72.8	0.505	3.38	31.0	41.6
MMR ($\lambda = 0.9$)	74.1	73.5	76.1	78.0	0.463	3.16	33.4	44.8
Maxmin	42.9	51.7	53.8	45.7	0.625	3.85	23.6	32.7
Greedy DPP	65.6	68.0	70.5	69.1	0.522	3.47	29.6	40.0
RNG-SCORE ($\gamma^* = 0.2$)	75.1	74.2	76.8	79.0	0.462	3.14	34.3	45.6

Table 3 extends the study to five BEIR tasks. The conclusion is uniform: every fixed-pressure diversifier loses on both NDCG@10 and α -NDCG@10 metrics. The tuned RNG-SCORE is the exception on the three seed-replicated tasks, where validation drives its margin into or near the fallback region and leaves it statistically indistinguishable from k -NN.

Table 3: NDCG@10 / α -NDCG@10 on five BEIR tasks (**bge-m3**, $k = 10$; MMR at $\lambda = 0.5$; best in bold, second best underlined). SciFact, FiQA and TREC-COVID are seed-replicated, with across-seed CIs up to 8 points on these small query sets. ArguAna and Touché are single-run.

Method	SciFact	FiQA	TREC-COVID	ArguAna	Touché
k -NN	64.1/64.1	41.7/46.7	66.1/70.2	38.3/38.3	25.2/27.8
MMR	52.9/52.9	29.6/33.0	37.0/40.3	0.2/0.2	10.9/12.9
Maxmin	49.4/49.4	27.2/30.2	46.7/52.0	0.9/0.9	9.2/11.7
Greedy DPP	56.0/56.0	30.9/34.4	39.8/43.8	1.0/1.0	11.0/13.2
RNG-SCORE	<u>63.4/63.5</u>	41.7/46.8	<u>65.6/69.6</u>	<u>26.4/26.4</u>	<u>17.8/19.6</u>

After a strong cross-encoder. Production pipelines rarely diversify a raw dense pool, they usually rerank first. We therefore repeat the test after the cross-encoder pipeline of Section 5.6. Table 4 reports the result on four datasets.

The cross-encoder only strengthens the negative result: CE top- k wins every comparison with the fixed-pressure diversifiers. The tuned RNG-SCORE tracks the cross-encoder ranking within a fraction of a point on every dataset, sitting at the CE top- k values and nominally above them on MuSiQue: validation lands its margin at or near one of the two fallback limits throughout (per-dataset margins in Appendix B). In every experiment of this subsection the annotation-free and intent-aware diversity metrics dissociate, diversifiers raising APD and Vendi while lowering α -NDCG and S-Recall.

6.2 RQ2: Manipulating pool redundancy

Misspecification regret. Table 5 reports, for each method, the best and worst S-Recall@5 over its hyperparameter grid on clean ($\rho = 0$) and heavily redundant ($\rho = 1$) pools and the resulting misspecification regret (5.1).

Three facts stand out. First, every fixed commitment carries regime risk, including the default: always- k -NN gives up 22.0 points in the redundant regime, the mirror image of MMR’s worst clean-pool toll (30.6 points). Second, the harm range of the fixed-pressure methods is set entirely by their operating point: against the k -NN rows of Tables 2 and 5, the $\lambda = 0.7$ that is optimal at $\rho = 1$ costs 5.9 points at $\rho = 0$ while the $\lambda = 0.3$ value that enforces a more diverse retrieval costs 30.6. Third, the relative-neighborhood family is the only method whose worst grid point never falls more than 3.9 points below the baseline in either regime.

Injected redundancy. Table 6 reports the sweep on HotpotQA with **bge-m3** over all 7,405 queries (three seeds, operating points tuned per ρ). At $\rho = 0$ the conclusions of Section 6.1 are reproduced. Injection then reverses the ordering: nearest-neighbor S-Recall stays essentially flat until the measured near-duplicate fraction

Table 4: Cross-encoder pipelines, $k = 5$ (MMR at its best $\lambda = 0.7$; CE+RNG-SCORE is the tuned semimetric variant of Appendix B; the non-HotpotQA datasets are single-run). Best per dataset and column in bold; a tie occupies both ranks, so cells below a tied best are not underlined.

Dataset	Method (post-CE)	Recall@5	NDCG@5	EM	F1
HotpotQA	CE top- k	89.6	87.9	41.0	53.3
	CE+MMR	84.6	83.4	38.2	50.0
	CE+DPP	80.0	80.8	36.2	47.7
	CE+RNG-SCORE	89.6	87.9	41.0	53.3
2WikiMultiHopQA	CE top- k	90.6	90.3	31.3	37.9
	CE+MMR	71.0	76.3	25.0	29.6
	CE+DPP	71.2	76.5	25.2	29.9
	CE+RNG-SCORE	90.6	90.3	31.3	37.9
MuSiQue	CE top- k	<u>74.3</u>	77.6	<u>26.0</u>	<u>34.1</u>
	CE+MMR	52.7	61.0	18.5	25.6
	CE+DPP	53.1	61.6	19.8	26.7
	CE+RNG-SCORE	74.4	77.6	26.3	34.3
NQ-Open	CE top- k	23.5	65.8	33.0	43.4
	CE+MMR	16.3	51.9	29.7	40.0
	CE+DPP	13.3	45.3	29.8	39.7
	CE+RNG-SCORE	23.5	65.8	33.0	43.4

Table 5: Misspecification regret on HotpotQA. Best and worst S-Recall@5 over each method’s grid on clean and heavily redundant pools and the worst shortfall against k -NN across both (for k -NN, against the best tuned method per regime). Lower regret is better. Best per column in bold, second best underlined.

Method	$\rho = 0$ (clean)		$\rho = 1$ (redundant)		max. regret
	best	worst	best	worst	
k -NN		<u>78.7</u>		52.6	22.0
MMR	78.0	48.1	74.6	53.5	30.6
Maxmin	45.7	45.7	50.6	50.6	33.0
Greedy DPP	69.1	69.1	<u>72.1</u>	72.1	<u>9.6</u>
RNG-SCORE	79.0	74.8	65.2	52.8	3.9

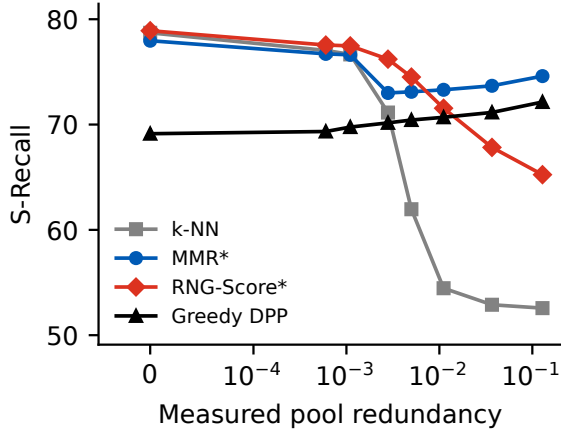
reaches approximately 0.001 (95% CI [0.0007, 0.0015] across seeds), then falls steeply as the Vendi score of the k -NN selection collapses from 3.1 toward 1.2. Past the crossover, both tuned diversifiers improve on k -NN by wide and highly significant margins and parameter-free DPP joins them from $\rho = 0.15$ (query-agnostic Maxmin remains below the baseline throughout).

Table 6: Redundancy-injection sweep, HotpotQA (bge-m3, $k = 5$); S-Recall@5, operating points tuned per ρ . Second header row: measured pool redundancy (4.1). Best per column in bold, second best underlined.

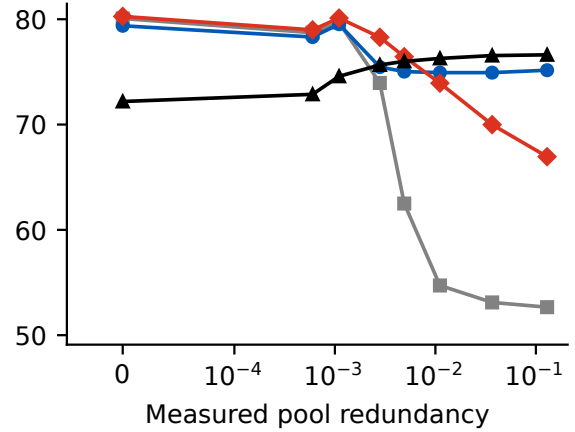
Method	$\rho=0$	0.025	0.05	0.1	0.15	0.25	0.5	1.0
	0	.0006	.0011	.0028	.0050	.011	.037	.129
k -NN	<u>78.7</u>	<u>77.0</u>	<u>76.7</u>	71.1	62.0	54.5	52.9	52.6
MMR (λ^*)	78.0	76.7	76.6	<u>73.0</u>	<u>73.1</u>	73.3	73.7	74.6
Maxmin	45.7	46.0	46.3	46.7	46.9	47.2	48.1	50.6
Greedy DPP	69.1	69.3	69.8	70.2	70.4	70.7	<u>71.2</u>	<u>72.1</u>
RNG-SCORE (γ^*)	78.9	77.6	77.5	76.2	74.5	<u>71.6</u>	67.8	65.2

The sweep further shows that no fixed operating point dominates the whole axis: the RNG-SCORE is the best method through $\rho = 0.15$, while MMR overtakes it from $\rho = 0.25$ onward yet gives up S-Recall on every clean level.

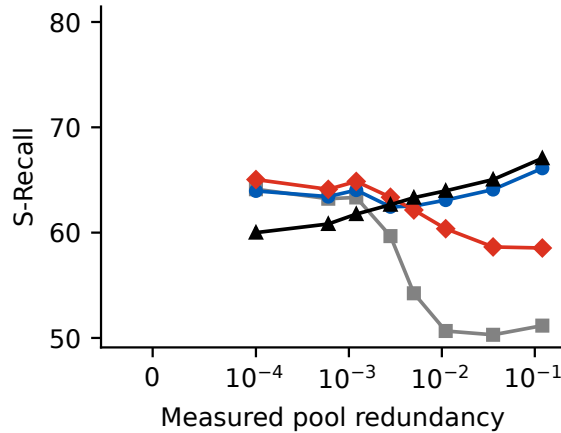
Figure 4 replicates the sweep under three encoders on the common measured-redundancy axis (encoder-invariant by construction on injection sweeps). The regime shape is fully reproduced under all three: the crossover location is shared only by the strong encoders, while the weak encoder crosses three times earlier (approximately 0.0003).



(a) bge-m3



(b) Qwen3-Embedding-4B



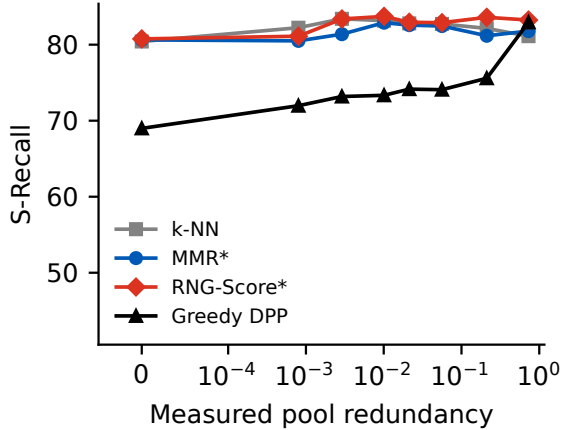
(c) all-MiniLM-L6-v2

Figure 4: The HotpotQA injection sweep under three encoders: S-Recall vs measured near-duplicate pair fraction (4.1), on common axes (log-scaled with the clean level kept on-axis).

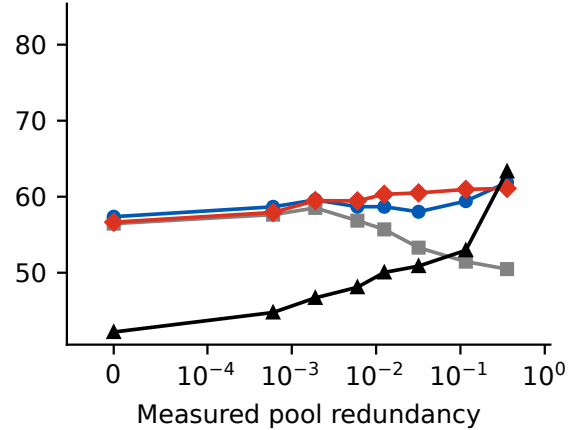
Evidence multiplicity on BEIR. Multi-evidence FiQA-2018 replicates the injection experiment on a natural non-QA benchmark (Figure 5b). The regime turns exactly as on multi-hop QA: both tuned diversifiers overtake k -NN significantly and stably from $\rho = 0.1$ onward, by up to +11.4 S-Recall@10 points at $\rho = 1$, with the tuned margin of the RNG-SCORE at an interior negative value across the sweep (Table 7). At $\rho = 0$ neither differs significantly from k -NN. Single-hop SciFact shows the opposite: under the same encoder, all relevance-aware methods yield essentially the same results (Figure 5a).

The tuned margin as an offline readout. Table 7 collects the validation-selected RNG-SCORE margins across the five injection sweeps. On the headline **bge-m3** sweep, the margin flips sign at the first injection level, +0.2 on the clean pool and negative in every seed thereafter. Under the other encoders, the flip arrives later, with seeds disagreeing in sign exactly in the transition zones. On SciFact, seeds disagree at nearly every level. On FiQA-2018 the margin sits at the interior value -0.2 at every level where the seeds agree, including the clean pool: FiQA’s natural pools are crowded at similarities well below the near-duplicate cutoff, the k -NN selection retaining an effective 3.5 of 10 distinct documents (Vendi) while the pair fraction (4.1) reads zero.

Natural redundancy through chunk overlap. Table 8 reports the chunk-overlap sweep of Section 5.6, where redundancy arises naturally from overlapping sliding windows: S-Recall is reported for every method



(a) SciFact (single-hop)



(b) FiQA-2018 (multi-hop)

Figure 5: The BEIR injection sweeps (**bge-m3**, $k = 10$, three seeds): S-Recall@10 vs measured near-duplicate pair fraction (4.1), on common axes. Single-evidence SciFact (a) stays flat across the sweep; multi-evidence FiQA-2018 (b) reproduces the multi-hop collapse-and-crossover pattern.

Table 7: Validation-selected RNG-SCORE margins γ^* per injection level: modal value across the three seeds where the seeds agree in sign, $\pm$ where they do not.

Sweep	$\rho=0$	0.025	0.05	0.1	0.15	0.25	0.5	1.0
HotpotQA, bge-m3	+0.2	-0.3	-0.3	-0.2	-0.2	-0.2	-0.2	-0.2
HotpotQA, Qwen3-Embedding-4B	+0.3	$\pm$	$\pm$	-0.2	-0.1	-0.1	-0.1	-0.1
HotpotQA, all-MiniLM-L6-v2	+0.3	+0.3	+0.2	+0.2	+0.2	$\pm$	-0.1	-0.1
SciFact, bge-m3	-0.2	$\pm$	$\pm$	-0.3	$\pm$	$\pm$	$\pm$	$\pm$
FiQA-2018, bge-m3	-0.2	$\pm$	-0.2	-0.2	-0.2	-0.2	-0.2	-0.2

both on the non-overlapping pool and at the heaviest overlap. On the multi-hop datasets, the regime appears, k -NN losing up to 9.5 points, while it barely moves on single-hop datasets. Overall, the relevance-aware diversifiers hold essentially flat across the sweep both for single-hop and multi-hop datasets. Moreover, with the exception of DPP on HotpotQA, they all score higher than k -NN in the more redundant setting.

The saturated-pool regime. A third point on the evidence-multiplicity axis is the *saturated* pool, where almost every candidate is relevant. TREC-COVID is the extreme case with a mean of 493 relevant documents per query. Under the chunk-overlap sweep (**bge-m3**, $k = 10$), k -NN coverage does not collapse, S-Recall@10 staying flat to slightly rising: there is always another relevant document to take a crowded slot. Yet, at the highest overlap, the tuned RNG-SCORE significantly leads k -NN (last row of Table 8). Saturation thus protects relevance-only selection from redundancy without eliminating the diversification gain, and the frozen rule of **RQ4** stays within noise of k -NN on the natural pools while gaining significantly at the heaviest overlap (Table 10).

All manipulations thus give the same reading of **RQ2**: raising pool redundancy does create a regime in which diversification pays, at a crossover that the deployment-level statistic (4.1) localizes on the measured axis, and the risk of committing to a method without knowing the regime is the misspecification regret of Table 5, up to 33 coverage points for the popular fixed commitments.

6.3 RQ3: Oracle headroom

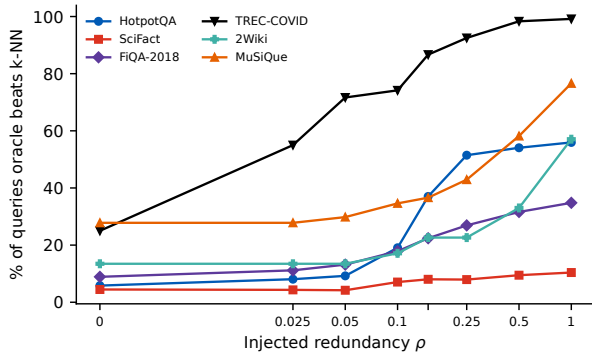
Results from the oracle protocol of Section 5.6 are reported in Figure 6. On clean pools of HotpotQA, the headroom is negligible: an oracle with ground-truth access that chooses freely among all methods and grid values per-query improves S-Recall over k -NN on only 5.8% of queries for a gain of 3.1 points. The headroom opens precisely at the crossover (a measured redundancy of 0.001, reached at $\rho = 0.05$), where the oracle beats

Table 8: Chunk-overlap sweep (**bge-m3**, $k = 5$; $k = 10$ for SciFact and TREC-COVID; 2Wiki abbreviates 2WikiMultiHopQA). Each cell reports the row method’s S-Recall@ k on the non-overlapping pool / at overlap 0.75 (tuned operating points per level); the header rows give each dataset’s hop count and measured redundancy (4.1) at overlap 0.75. Best per dataset in bold, second best underlined (ties both bold and occupying both ranks). The NQ-Open sweep is single-run; across-seed CIs reach 1.1 points on the QA columns and 6.4 points on the small SciFact and TREC-COVID query sets; saturated TREC-COVID (mean 493 relevant documents per query) has no hop count.

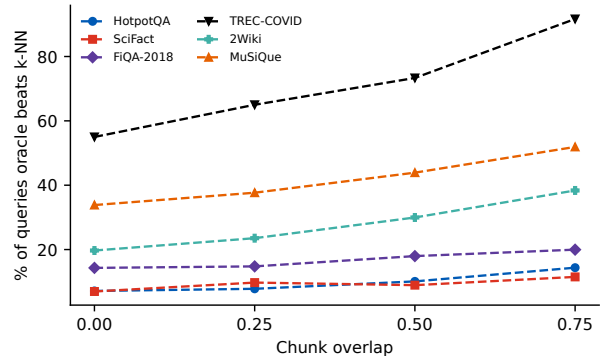
Method	HotpotQA	2Wiki	MuSiQue	SciFact	NQ-Open	TREC-COVID
hops	2	2	2–4	1	1	sat.
red. ₇₅	.001	.005	.005	.002	.005	.006
k -NN	77.8/74.7	85.7/77.0	69.5/60.0	76.8 /73.0	92.1/89.7	21.8 /23.5
Maxmin	46.2/47.9	69.3/66.8	48.3/46.4	57.7/62.2	86.8/87.6	15.0/22.2
Greedy DPP	69.4/70.7	84.4/ 85.1	<u>71.5</u> / 71.9	72.9/74.3	91.6/92.0	18.6/26.2
MMR *	77.2/ <u>76.3</u>	85.5/80.8	<u>71.5</u> /70.8	76.8 / 75.6	<u>92.8</u> / 93.0	<u>21.3</u> / <u>27.2</u>
RNG-SCORE *	77.9 / 77.9	85.9 / <u>83.9</u>	72.0 / <u>71.7</u>	75.6/ <u>75.4</u>	92.9 / 93.0	21.1/ 27.3

k -NN on 9.2% of queries, and peaks when redundancy is maximal, where it improves on k -NN on over half of queries (its per-level coverage appears as the oracle row of Table 9). On single-hop SciFact, it stays at 4–10% across the whole sweep. On multi-hop FiQA-2018, whose crowded natural pools already leave the oracle 8.9% of queries to improve on the clean level, the headroom climbs monotonically to 34.8% at $\rho = 1$. On saturated TREC-COVID, the injection sweep confirms the chunk-overlap picture: the oracle already beats k -NN on a quarter of the queries on the natural pools and on 55–99% across the injection levels.

The chunk-overlap sweeps reproduce the pattern (Figure 6b): headroom rises with overlap on every multi-evidence dataset, stays low on the single-hop datasets (5–7% on NQ-Open, 7–12% on SciFact) and is largest on saturated TREC-COVID (55–92%), where a better selection almost always exists even though k -NN never collapses.



(a) Injection sweeps



(b) Chunk-overlap sweeps

Figure 6: Oracle headroom: percentage of test queries on which the all-methods oracle beats k -NN (HotpotQA, 2Wiki and MuSiQue at $k = 5$; SciFact, FiQA-2018 and TREC-COVID at $k = 10$; three seeds throughout). (a) Injection sweeps vs injected fraction ρ (log-scaled axis with the clean level kept on-axis); (b) Chunk-overlap sweeps vs overlap.

6.4 RQ4: The decision rule

Under the tuning protocol of Section 5.6, applied once to the **bge-m3** HotpotQA injection sweep with the fallback diversifier ranging over every method and grid value, free threshold tuning selects $\tau^* = 1.83 \pm 0.31$ across that sweep’s seeds with $D^* = \text{MMR}(\lambda = 0.7)$ in every seed. Retuned on the $k = 10$ replica of the same sweep, it selects $\tau^* = 1.9$ (1.9/1.9/1.8 per seed). The tuned threshold thus stays at the evidence requirement $h = 2$ for both $k = 5$ and $k = 10$. The selection is insensitive: every threshold in $\tau \in [1.5, 2.5]$ performs within half a point of the optimum at $k = 5$ (Figure 7a) and replacing the Vendi trigger with the mean pairwise distance of the k -NN selection reproduces the headroom capture within half a point at every level. We therefore freeze the deployed rule at $\tau = h$, that is 2 on the two-hop benchmarks, and use it everywhere below. The

threshold is interpretable: we diversify only when the selection carries fewer effectively distinct documents than the query needs.

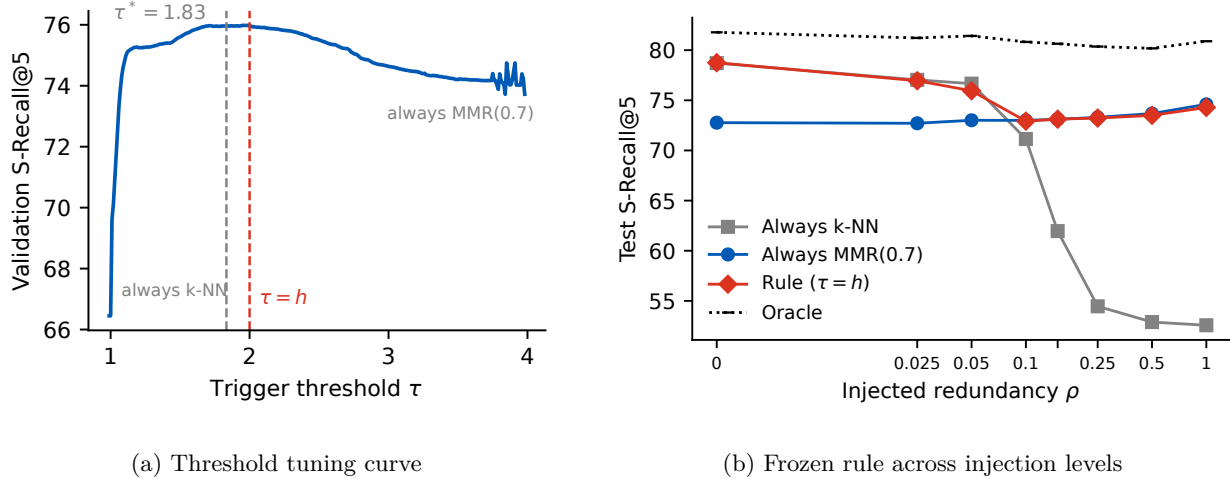


Figure 7: The decision rule on the HotpotQA sweep. (a) Pooled validation S-Recall@5 as the trigger threshold τ varies continuously. (b) The frozen rule per injection level (log-scaled ρ axis, ticks at the sweep levels).

Table 9: The frozen rule ($\tau = h = 2$, $D^* = \text{MMR}(0.7)$) per injection level, S-Recall@5. “Per-level tuned” knows the regime; the oracle is the all-methods per-query bound of Section 6.3; the trigger rate is the fraction of queries on which the rule diversifies. Among the three policies, best in bold, second best underlined.

Selector	$\rho=0$	0.025	0.05	0.1	0.15	0.25	0.5	1.0	pool
always k -NN	78.7	77.0	76.7	71.1	62.0	54.5	52.9	52.6	65.7
always D^*	72.8	72.7	73.0	73.0	73.1	73.3	73.7	74.6	<u>73.3</u>
rule	78.7	<u>76.9</u>	<u>76.0</u>	<u>72.9</u>	73.1	<u>73.2</u>	<u>73.5</u>	<u>74.3</u>	74.8
trigger %	0.7	3.3	15.5	81.0	87.8	90.0	91.4	92.9	
<i>Reference bounds (not deployable policies)</i>									
per-level tuned	78.9	77.6	77.5	76.2	74.5	73.3	73.7	74.6	75.8
oracle	81.8	81.2	81.4	80.8	80.6	80.4	80.2	80.9	80.9

Table 9 is the central practical result of the paper. The frozen rule tracks the upper envelope of the two fixed policies across the whole axis. On the three cleanest levels, it fires on under 16% of queries and stays within 0.7 points of k -NN, avoiding the toll the diversifier pays there; from $\rho = 0.1$ onward it fires on 81–93% and lands within 0.3 points of always-MMR. Pooled across levels it captures 60% of the all-methods oracle headroom, a bound that uses ground-truth labels per query, and 91% of the gain of the per-level tuned selector.

Transfer of the frozen rule. We apply the frozen rule with no retuning to fifteen further fine-grid sweeps, grouped in Table 10 by what they change. All use $k = 5$ except the HotpotQA $k = 10$ injection replica and the SciFact, FiQA-2018 and TREC-COVID sweeps. The threshold everywhere is the task’s evidence requirement of Section 5.6. The rule transfers across encoders, across k , across the redundancy mechanism and across datasets. On clean pools, it either gains or stays within noise of k -NN on all but two sweeps, namely the FiQA-2018 and TREC-COVID injection sweeps. Under redundancy, the rule gains on every multi-evidence sweep except on the chunk overlap run of FiQA-2018, where it stays within noise of k -NN. On single-hop NQ-Open the trigger condition is unsatisfiable and the rule is exactly k -NN, while on saturated TREC-COVID, whose relevant-set sizes keep the trigger firing on every query, the rule coincides with always-MMR (0.7): under chunk overlap its nominal loss on the non-overlapping pool is within noise of k -NN and its gain at the heaviest overlap is significant, while under injection it trades a small clean-pool toll for one of the largest heavy-redundancy gains in the table.

Table 10: Transfer of the frozen rule ($\tau = h$, $D^* = \text{MMR}(0.7)$) to further sweeps with no retuning, grouped by what each sweep changes relative to the tuning sweep (**bge-m3** HotpotQA injection). Each Δ is the rule’s S-Recall@ k minus k -NN’s at the same sweep level, averaged over seeds (positive favors the rule): **clean** Δ is the worst Δ over the near-clean levels ($\rho \leq 0.05$ on injection sweeps, overlap ≤ 0.25 on chunking sweeps), **heavy** Δ the Δ at the heaviest level ($\rho = 1$, respectively overlap 0.75) and **pooled** Δ the difference with all levels pooled.

Transfer target	clean Δ	heavy Δ	pooled Δ
<i>Different encoder (injection)</i>			
Qwen3-Embedding-4B	−0.3	+22.5	+9.8
all-MiniLM-L6-v2	−0.2	+13.6	+5.9
<i>Different k (injection)</i>			
HotpotQA ($k=10$)	0.0	+24.1	+5.7
<i>Different mechanism (chunk overlap)</i>			
HotpotQA	0.0	+1.4	+0.4
<i>Different dataset (injection)</i>			
SciFact (single-hop, $k=10$)	−0.7	+1.8	+0.4
FiQA-2018 (multi-hop, $k=10$)	−1.0	+9.2	+2.9
2WikiMultiHopQA (multi-hop)	0.0	+10.7	+1.5
MuSiQue (multi-hop)	+0.7	+21.5	+5.6
TREC-COVID (saturated, $k=10$)	−1.7	+12.9	+3.1
<i>Different dataset and mechanism (chunk overlap)</i>			
NQ-Open (single-hop)	0.0	0.0	0.0
SciFact (single-hop, $k=10$)	−0.4	+0.4	0.0
FiQA-2018 (multi-hop, $k=10$)	−1.0	+0.3	−0.2
2WikiMultiHopQA (multi-hop)	0.0	+3.2	+1.0
MuSiQue (multi-hop)	+2.3	+10.0	+5.4
TREC-COVID (saturated, $k=10$)	−2.0	+4.1	+0.7

6.5 RQ5: Answer quality under redundancy

We re-run the injection sweep on HotpotQA and 2WikiMultiHopQA under the generation protocol of Section 5.6. We see in Figure 8 that the picture established at the retrieval level carries over to answer quality. On clean pools diversification hurts, MMR costing a significant 3.1 EM points against k -NN. Under redundancy, k -NN answer quality collapses while the diversifier holds flat. Moreover, as previously observed at the retrieval level, the decision rule tracks the upper envelope, sitting on k -NN at $\rho = 0$ (firing on under 1% of queries), switching to the diversifier once redundancy appears and staying within 0.2 EM points of the better of the two policies at every level (Table 11). 2WikiMultiHopQA tells a softer but similar story: the diversifier’s and the rule’s advantage materializes only at the highest level, the rule staying within 0.7 EM points of k -NN at moderate ρ .

Table 11: Answer quality (EM/F1) at the clean and heaviest injection levels, generator **flan-t5-base** (a single run per dataset). Best in bold, second best underlined.

Method	HotpotQA		2WikiMultiHopQA	
	$\rho=0$	$\rho=1$	$\rho=0$	$\rho=1$
k -NN	34.0/45.2	26.5/36.0	31.4/38.1	27.4/32.6
MMR ($\lambda=0.7$)	30.8/41.5	31.2/42.1	29.7/35.8	<u>29.4/35.4</u>
rule ($\tau=h$)	34.0/45.2	<u>31.0/41.8</u>	31.4/38.1	29.5/35.4

7 Discussion

7.1 Reconciling the literature

Read jointly, the results of Section 6 resolve the contradiction of Section 2. Studies reporting that diversity benefits retrieval in RAG pipelines operate on redundant pools: chunked long contexts [37], iterative pipelines that re-fetch overlapping content [52], redundancy-heavy benchmarks [17, 48] or older first-stage encoders of the fixed-pool selection methods [17, 41]. AdaGReS makes the dependence to redundancy explicit by targeting near-duplicate chunks [47]. Studies reporting that diversity does not improve retrieval, including the controlled

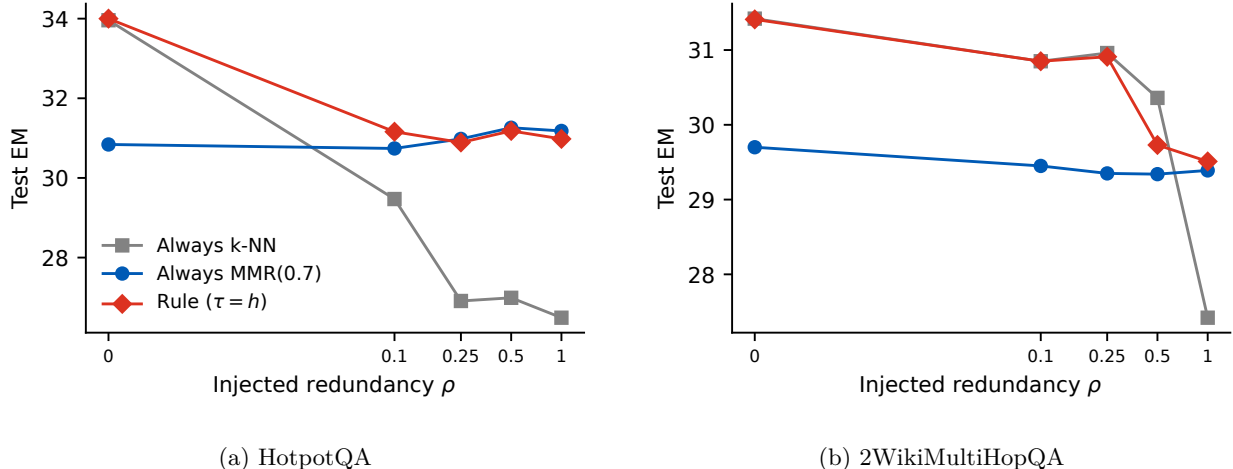


Figure 8: Exact match vs injected redundancy on (a) HotpotQA and (b) 2WikiMultiHopQA. Log-scaled ρ axis with ticks at the sweep levels.

evaluations of Eibich et al. [21] and Bal and Puhon [6], operate on nonredundant pools. The cross-encoder comparison of Section 6.1 likewise reproduces the finding that reranking, not diversification, is the high-leverage stage of the pipeline [71]. Neither camp is wrong about its own regime: the error is the implicit claim that the conclusion transfers across regimes.

The reported sweeps quantify how low the boundary between regimes sits on the strong-encoder multi-hop tasks. A measured near-duplicate pair fraction of approximately 0.001 is enough for k -NN retrieval to collapse and the use of a weaker encoder moves that boundary by half an order of magnitude (see Figure 4c), which is how the evaluations using older encoders land in the regime where diversification pays. The oracle bounds explain the adaptive end of the literature the same way: adaptivity, like diversification itself, is valuable in proportion to measured redundancy, the per-query headroom moving from about 6% of queries to over half across the crossover (Figure 6), which is why per-query selection of the diversity level helps in the chunked long-context setting of Khan et al. [37] yet has nothing to capture on many benchmarks.

Other works in the literature reporting positive results regarding retrieval diversification concern notions of diversity different from the one studied here: diversity on the query side, whether through lexical decomposition [84] or coverage of a sub-query demand distribution [82], selection for long contexts [72] and open-ended tasks in which answer diversity is itself rewarded [27, 54]. None of these evaluates geometric diversification of a fixed candidate pool where diversity is merely instrumental, so their conclusions complement rather than contradict the present analysis.

7.2 Clean pools resist diversification

The systematic dissociation between annotation-free and intent-aware diversity metrics is, after the decision rule, the most practically consequential observation of the study. APD and the Vendi score are routinely reported as evidence that a diversified selection is better. However, on every dataset we evaluate, methods that maximize them minimize α -NDCG, S-Recall, and answer quality. Our interpretation is geometric: in modern embedding spaces, the distinct facets of a multi-hop question occupy a compact neighborhood of the query, so pushing a selection out of that region substitutes gold passages for related but irrelevant ones. These are harmful for the generator [3, 19] and add nothing to the context sufficiency that generation actually requires [34]. ArguAna is the extreme case: the single relevant counter-argument sits directly adjacent to the query in embedding space, so any outward pressure discards it, which is how MMR collapses to an NDCG@10 of 0.2 (Table 3). The all-methods oracle bound of Section 6.3 allows us to push the analysis further: on nonredundant pools the problem is not that the diversifiers are badly tuned or insufficiently adaptive, but that there is almost nothing for diversification to gain over k -NN. Thus, evaluations should treat geometric spread as a descriptive statistic rather than an objective and should instead report coverage-aware metrics whenever labels permit. Those spread statistics may nonetheless be used as diagnostics, analogously to the decision rule using the Vendi

score as a redundancy gauge.

7.3 Reading the regime from the RNG-Score

The margin trajectories obtained from the RNG-SCORE in Table 7 are the hinge geometry at work. A negative margin restricts the obstruction penalty to pairs at duplicate distance scale, demoting copies while leaving distinct evidence untouched, while a positive margin penalizes more severely distinct documents, meaning that there is no possible gain to be made with near-duplicates. The margin thus reads duplicate-scale structure specifically, not the regime at large, and the other trajectories confirm it: a sign flip occurs at different levels for different encoders because each geometry places duplicate pairs at a distinct scale relative to query–document distances. The weak encoder stays positive longest, its low-redundancy gains coming from a more conflated structure rather than from literal duplicates. Furthermore, seeds disagree in sign exactly where the validation objective is flat, i.e. in the transition zones. On SciFact however, they disagree across nearly the whole axis, meaning that the margin is reading noise precisely where there is nothing to measure. Therefore, we see that a small negative margin signals duplicate structure worth removing, while a fallback-region or seed-unstable margin indicates a pool with nothing to remove.

7.4 Deploying the rule

The sweeps show that no fixed method dominates the redundancy axis, so a deployed policy should adapt accordingly. The decision rule proposed here adapts on the right variable: its trigger reads off exactly the quantity whose collapse causes the k -NN failure and the per-query distributions of $T(q)$ barely overlap across the crossover, which is why a wide band of thresholds performs identically and why the frozen rule transfers to redundancy levels and mechanisms it never saw. Adapting per query matters because the crossover location itself is specific to each task. Evidence multiplicity sets how steeply the regime turns: gold-concentrated redundancy can crowd out a needed document only when the query needs more than one, which is why the multi-hop datasets and multi-evidence FiQA collapse at a few near-duplicate per mille while single-hop SciFact and NQ-Open reach the regime only under extreme duplication, mildly even then. Across our experiments, the crossover spans at least 0.0003 (the weak encoder) to approximately 0.006 (FiQA at $k = 10$) and does not exist on SciFact. Thus, although the specific location does not generalize, the existence and sharpness of a measurable crossover does on multi-hop datasets and the trigger localizes that boundary per query without ever measuring it.

Evidence multiplicity also anchors the rule’s threshold. The two variables of the analysis play the distinct roles that the rule’s architecture mirrors: redundancy is the lever, the dynamic variable that is usually unmeasured in production pipelines; while evidence multiplicity supplies the threshold, treated in this paper as a static workload property because our evaluation reads h from gold annotations, the hop count on multi-hop QA and the relevant-set size on the graded-relevance tasks, quantities that a deployment does not have per query. Label-free per-query estimators nevertheless exist off the shelf, such as query-complexity classifiers routing queries by predicted retrieval depth [32] and query decomposition counting a question’s sub-questions directly [49]. Thus, the practical cost of fixing $\tau = h$ is a single integer either estimated per query or for a whole workload. A natural refinement, left to future work, is therefore to supply the threshold with the query’s own estimated evidence requirement, $T(q) < \hat{h}(q)$ with $\hat{h}(q)$ obtained from one of these estimators. This would be another step forward since, to our knowledge, no existing adaptive pipeline couples a diversification decision with an estimated hop count, the adjacent adaptive mechanisms selecting instead how many documents to keep per query [45, 78] or diversifying across the inferred interpretations of ambiguous questions [28].

The rule is the instrumental-credit counterpart of selective diversification [57], which gates web-search diversification on predicted query ambiguity [18]. Among the adaptive pipelines of Khan et al. [37], Rezaei and Dieng [52] and Stouras et al. [63], it is the minimal member, spending an eigendecomposition per query where those spend planner and evaluator LLM calls or a trained router. On clean pools, the all-methods oracle bound implies that no adaptive policy, however expensive, can do appreciably better, and on redundant pools the rule’s remaining gap to the oracle bounds what richer policies can still contribute.

7.5 Bounded regret as a design requirement

The regret analysis generalizes beyond the methods compared. A fixed diversification strength interacts unpredictably with pool geometry, as MMR’s collapse on ArguAna illustrates (Table 3), so bounded regret should be a design requirement for any diversifier, met structurally (the relative-neighborhood scores’ dual fallback limits) or operationally (gating an unbounded-regret diversifier behind a redundancy trigger). The two routes are complementary, the gated rule achieving the better headline numbers and the bounded score supplying an interpretable margin.

8 Limitations

Six limitations qualify the conclusions. None, in our assessment, threatens the regime analysis itself. (i) *Statistical coverage*. The injection sweeps, the chunk-overlap sweeps other than NQ-Open’s, the oracle and decision-rule analyses, the seed-replicated BEIR comparisons and the HotpotQA cross-encoder comparison carry three seeds with across-seed confidence intervals. The generation comparisons (Section 6.5), the two argument-retrieval BEIR tasks, the remaining cross-encoder datasets and the NQ-Open chunking sweep are single runs with per-query paired tests only. (ii) *Redundancy realism*. Redundancy was simulated using sentence-shuffled copies and natural chunk-overlap protocols, but other methods such as injecting paraphrases remain untested, as does the sensitivity of the crossover location to the injection protocol. The chunking sweeps also score a subtopic as covered when any window of a gold passage is selected, an optimistic convention (a sixty-word window need not contain the answer-bearing content) that the chunking protocol does not yet pair with generation. (iii) *Scope of the rule and its guarantees*. The rule’s threshold is the task’s evidence requirement, which in deployment requires task-specific knowledge of evidence multiplicity rather than per-query labels, and the rule’s behavior under estimated rather than annotated h is untested. The $k = 10$ transfer evidence is limited to the HotpotQA injection replica and the SciFact, FiQA-2018 and TREC-COVID sweeps. Margin tuning inherits validation-split noise, so an adequate tuning set is required (on TREC-COVID, which only contains fifty queries, the selected margin varies across splits). Moreover, the oracle applies only to policies that adapt a scalar or the method per query, not policies that change the pool or the query. (iv) *Task and domain scope*. The study covers English factoid, multi-hop QA and BEIR-style retrieval. Open-ended information seeking, where comprehensiveness or answer diversity gives diversity independent value [27, 54], is out of scope and our conclusions do not extend to it. (v) *Generation*. Because the instrumental-credit argument ultimately rests on answer quality, the conclusions of the paper should be confirmed with a modern and larger instruction-tuned generator, which may handle distractors and long redundant contexts differently. (vi) *Baselines*. The diversifiers compared are the classical fixed-pressure methods: isolating recent adaptive and Vendi-objective selectors on the same fixed pools is left to future work.

9 Conclusion

This paper asked when the retrieval stage of a RAG pipeline should diversify and answered with a regime analysis organized around a main observable variable: whether the candidate pool contains redundancy. On clean pools, the answer on the task families studied here is: essentially never. In most cases, the fixed-pressure diversifiers significantly degrade relevance, coverage-aware diversity and answer quality. The geometric spread that diversifiers deliver is anti-correlated with the coverage they promise and the all-methods per-query oracle certifies that only about 6% of queries admit any diversification gain at all. The regime changes when the pool itself is redundant, whether by injection or by overlapping chunking. Then, every relevance-aware diversifier overtakes nearest-neighbor selection by a wide margin.

The practical contribution is that the regime need not be guessed. On the multi-hop tasks, the crossover occurs at a measured near-duplicate pair fraction far below what casual inspection would flag, around 0.001 under both strong encoders and the natural chunking mechanism, although the location itself is specific to each dataset. In addition, a label-free statistic that any retrieval system can compute, the Vendi score of its k nearest neighbors, detects the regime per query, so the crossover location is not needed to deploy the rule. The proposed decision rule, i.e. diversify when that score falls below the number of evidence pieces h the query requires, does virtually no harm on clean pools, recovers most of the value of perfect regime knowledge on redundant ones and transfers with no retuning across encoders, datasets and k . When the threshold is left free,

validation recovers a large plateau of possible values which includes h , namely the evidence multiplicity, which is a property of the workload a deployment knows or should estimate in practice.

The analysis also yields an evaluation standard and an instrument: misspecification regret, which across the classical diversifiers and the k -NN default spans 10–33 coverage points, and the RNG-SCORE, whose dual nearest-neighbor limits bound that regret by construction and whose tuned margin flips sign exactly when the pool acquires duplicate structure.

Code and data availability. The full evaluation pipeline (sweep scripts, derived analyses, decision-rule and transfer tooling, and the per-run result files behind every table and figure) will be released upon publication.

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A Theoretical properties

We present here the facts summarized in Section 3.3. Recall that for a candidate pool $V \subset \mathbb{R}^D$, a candidate $w \in V$, a query $q \in \mathbb{R}^D$ and a parameter $\gamma \in \mathbb{R}$, the RNG-SCORE is defined as

$$\text{score}_\gamma(w; q, V) = d(q, w) + \sum_{v \in P(w; q, V)} [d(q, w) - d(v, w) + \gamma]_+,$$

where $P(w; q, V) = \{v \in V : d(q, v) < d(q, w)\}$ is the set of closer competitors of w .

Proposition A.1 (Exact recovery of the hard admissibility baseline). *For any $w \in V$ and any $\gamma \in \mathbb{R}$, $\text{score}_\gamma(w; q, V) = d(q, w)$ if and only if every closer competitor $v \in P(w; q, V)$ satisfies $d(v, w) \geq d(q, w) + \gamma$.*

Proof. Every summand in the RNG-SCORE is nonnegative, so $\text{score}_\gamma(w; q, V) = d(q, w)$ exactly when every hinge term is zero. The hinge for v is zero iff $d(q, w) - d(v, w) + \gamma \leq 0$, i.e. $d(v, w) \geq d(q, w) + \gamma$. $\square$

Proposition A.2 (Monotonicity in γ). *For every fixed $w \in V$, the map $\gamma \mapsto \text{score}_\gamma(w; q, V)$ is nondecreasing.*

Proof. Each term $[d(q, w) - d(v, w) + \gamma]_+$ is nondecreasing in γ and a finite sum of nondecreasing functions is nondecreasing. $\square$

Proposition A.3 (Similarity invariance). *Let d and $\tilde{d}$ be semimetrics on X and $\tilde{X}$ respectively and suppose $f : X \rightarrow \tilde{X}$ satisfies $\tilde{d}(f(x), f(y)) = \lambda d(x, y)$ for all $x, y \in X$ and some $\lambda > 0$. Then, writing score_γ^d for the RNG-SCORE under d ,*

$$\text{score}_{\lambda\gamma}^{\tilde{d}}(f(w); f(q), f(V)) = \lambda \text{score}_\gamma^d(w; q, V)$$

for every finite $V \subset X$ and $w \in V$. Hence, isometries preserve the score while similarities preserve the induced ordering.

Proof. Since $\lambda > 0$, we see that $d(q, v) < d(q, w)$ iff $\tilde{d}(f(q), f(v)) < \tilde{d}(f(q), f(w))$, so the closer-competitor set of $f(w)$ is $f(P(w; q, V))$. Also, every term in the score is scaled by λ and since $[\lambda x]_+ = \lambda [x]_+$, we deduce that $\text{score}_{\lambda\gamma}^{\tilde{d}}(f(w); f(q), f(V))$ is simply a scaling of $\text{score}_\gamma^d(w; q, V)$ by factor λ . $\square$

Proposition A.4 (Nearest-neighbor limit for $\gamma \rightarrow -\infty$). *If V is finite, there exists $\gamma_0 \in \mathbb{R}$ such that for every $\gamma \leq \gamma_0$, we have $\text{score}_\gamma(w; q, V) = d(q, w)$ for all $w \in V$. Thus, ranking by RNG-SCORE then coincides with ranking by distance to the query.*

Proof. The set $\{d(v, w) - d(q, w) : w \in V, v \in P(w; q, V)\}$ is finite; let γ_0 be its minimum. If $\gamma \leq \gamma_0$, every admissible pair has $d(q, w) - d(v, w) + \gamma \leq 0$, so every hinge term vanishes. $\square$

Proposition A.5 (Nearest-neighbor limit for $\gamma \rightarrow +\infty$). *Suppose V is finite and in general position with respect to q , meaning that the values $\{d(q, v)\}$ are all distinct. For sufficiently large γ , the ranking induced by RNG-SCORE coincides with the k -NN ranking.*

Proof. As $\gamma \rightarrow +\infty$, every hinge term becomes active and equals $[d(q, w) - d(v, w) + \gamma]_+ = \gamma + O(1)$. Thus, for large enough γ , we have $\text{score}_\gamma(w; q, V) = n(w)\gamma + O(1)$ where $n(w) = |P(w; q, V)|$. Since the distances $d(q, w)$ are distinct, $w \mapsto n(w)$ is a bijection onto $\{0, \dots, |V| - 1\}$ and ranking by $n(w)$ is equivalent to ranking by $d(q, w)$. $\square$

Corollary A.6 (Fallback guarantee under validated tuning). *Let $A \subset \mathbb{R}$ be a finite tuning grid containing some $\gamma \leq \gamma_0$ (Proposition A.4) and let γ^* maximize any evaluation functional of the induced top- k ranking over a validation set. Then the validation performance at γ^* is at least that of nearest-neighbor ranking, which is realised by a grid member.*

B Cross-encoder semimetric

This appendix records how the RNG-SCORE, defined through an embedding-space distance, is adapted to the cross-encoder setting of Section 6.1.

Recall that a *semimetric* on a set X is a function $d : X \times X \rightarrow \mathbb{R}$ satisfying, for all $x, y \in X$: $d(x, x) = 0$; $d(x, y) > 0$ when $x \neq y$; and $d(x, y) = d(y, x)$. A metric additionally satisfies the triangle inequality $d(x, z) \leq d(x, y) + d(y, z)$.

In what follows, $\text{sim}_{\text{CE}}(q, w) \in [0, 1]$ denotes the cross-encoder relevance score of candidate w for query q , and d denotes the underlying embedding-space distance (used only for document–document comparisons).

Essentially, the construction defines a semimetric d_{CE} on $V \cup \{q\}$ and applies the RNG-SCORE directly: set $d_{\text{CE}}(q, q) = 0$ and $d_{\text{CE}}(q, v) = d_{\text{CE}}(v, q) = 1 - \text{sim}_{\text{CE}}(q, v)$ for all $v \in V$, so that

$$P(w; q, V) = \{v \in V : \text{sim}_{\text{CE}}(q, v) > \text{sim}_{\text{CE}}(q, w)\},$$

i.e. the closer competitors of w are exactly the candidates the cross-encoder ranks above it. This is an instance of the decoupled-geometry observation of Section 3.3: relevance ordering from the cross-encoder, redundancy from the embedding space. The document–document side requires making the two scales commensurable per pool. The reference choice is the linear normalization

$$d_{\text{CE}}(v, w) = \frac{d(v, w)}{R}, \quad R = \max_{v \in V} d(q, v),$$

giving

$$\text{score}_\gamma(w; q, V) = 1 - \text{sim}_{\text{CE}}(q, w) + \sum_{v \in P(w; q, V)} \left[1 - \text{sim}_{\text{CE}}(q, w) - \frac{d(v, w)}{R} + \gamma \right]_+.$$

In the experiments, four alternative embedding distance variants (sigmoidal and reciprocal normalizations of $d(v, w)$, a relevance-scaled distance and the raw cosine distance) were considered as well. All five variants land within half an F1 point of one another at their tuned margins, so nothing of substance hinges on the choice.

In the cross-encoder experiments of Section 6.1, the validation-tuned margins land at or near the fallback limits on every dataset: $\gamma^* = -0.5$ on HotpotQA, -0.3 on 2WikiMultiHopQA (where Proposition A.4 deactivates the penalty and the cross-encoder ranking is reproduced exactly), -0.35 on NQ-Open and $+1.1$ on MuSiQue, near the positive limit of Proposition A.5.

C Kernelization

For any positive-semidefinite kernel K on $V \cup \{q\}$, the kernel-induced squared distance $d_K(x, y) = K(x, x) + K(y, y) - 2K(x, y)$ is a semimetric, which is all the RNG-SCORE requires, and every hinge term depends on kernel evaluations only so the score never needs explicit coordinates. Since a DPP consumes the same input, one kernel feeds both methods: DPP performs global log-determinant inference, the RNG-SCORE local obstruction scoring. The binary lune condition is invariant under squaring ($x < y \Leftrightarrow x^2 < y^2$ for nonnegative reals), so kernelization changes only the soft penalty weighting, not the obstruction graph. For unit-normalized embeddings, the linear-kernel d_K is twice the cosine distance, so the kernel view adds content exactly where vector norms carry information.

D Single-stage comparisons on the remaining QA datasets

Table 12 reports single-stage reranking on the non-overlapping pools of the three QA datasets whose main-body comparisons run under the cross-encoder pipeline (Table 4), alongside each pool’s measured redundancy (4.1). 2WikiMultiHopQA reproduces the HotpotQA conclusions: every fixed-pressure diversifier is significantly below k -NN and the tuned RNG-SCORE sits marginally above it. Mild diversification pays on MuSiQue: +2.1 S-Recall points for parameter-free greedy DPP and for MMR ($\lambda = 0.9$) while +2.5 for the tuned RNG-SCORE. NQ-Open sits between, with small gains for the mildest operating points.

The tuned γ margins are read as in Section 7 jointly with how often the induced selection deviates from k -NN. On 2WikiMultiHopQA, the selected $\gamma = -0.3$ is the fallback end of the sweep grid and reproduces k -NN on 98.8% of queries: validation essentially says “do not diversify”. On NQ-Open, a single-hop dataset, the mild negative margin acts on only 2.8% of queries and, where it acts, improves S-Recall on twice as many queries as it degrades. On MuSiQue, the selected margin is mild ($\gamma = 0.1$, with $\gamma = -0.1$ performing indistinguishably). The penalty fires at ordinary inter-passage distances rather than at duplicate scale, deviating from k -NN on 16.5% of queries. What that pressure buys tracks evidence multiplicity rather than duplicate structure: split by hop count, the tuned margin’s gain grows from +1.1 S-Recall points on 2-hop queries through +3.3 on 3-hop to +4.8 on 4-hop, where the k -NN baseline falls to 50.5 as up to four gold paragraphs compete for five slots. MMR ($\lambda = 0.9$) and parameter-free DPP show the same gradient, and the 2-hop rows behave like 2WikiMultiHopQA.

Table 12: Single-stage reranking on the natural pools of the remaining QA datasets (S-Recall@5, **bge-m3**, single run each; red_0 = measured pool redundancy (4.1); tuned operating points $\lambda = 0.9/0.9/0.7$ and $\gamma = -0.3/0.1/-0.1$ for 2WikiMultiHopQA/MuSiQue/NQ-Open). Best in bold, second best underlined.

Dataset	red_0	k -NN	Maxmin	DPP	MMR *	RNG-Score *
2WikiMultiHopQA	0.0006	<u>85.7</u>	69.3	84.4	85.5	85.8
MuSiQue	0.0011	69.2	48.4	<u>71.3</u>	<u>71.3</u>	71.7
NQ-Open	0.0007	92.1	86.8	91.6	<u>92.8</u>	92.9